

Divisibility of central binomial coefficients

OPG-506

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Problem (1) Prove that there exist infinitely many positive integers n such that

$$\gcd\left(\binom{2n}{n}, 3 \cdot 5 \cdot 7\right) = 1.$$

Problem (2) Prove that there exists only a finite number of positive integers n such that

$$\gcd\left(\binom{2n}{n}, 3 \cdot 5 \cdot 7 \cdot 11\right) = 1.$$